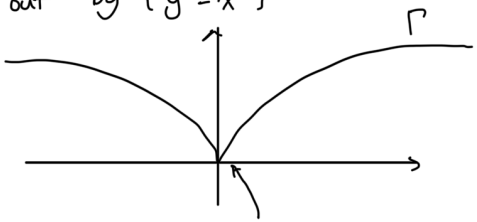
Example. (The cuspidal cubic)

The cuspidal cubic is the curve in $\mathbb{R}^2$ traced out by $\{y^3 = x^2\}$



this curve has one singular point at the origin

Question. Can we detect this algebraically?

Answer: The existence of the singular point is detected by the fact that "not all" polynomials on Γ are actually polynomials.

↳ In other words, the kink at the origin is able to obfuscate or hide bad functions.

For instance, consider the fcn $\frac{x}{y}$.

This is clearly not a polynomial on $\mathbb{R}^3$, However, on the graph Γ, notice that

$$\left(\frac{x}{y}\right)^3 = \frac{x^3}{y^3} \equiv \frac{x^3}{x^2} = x.$$

$(\Gamma: y^2 = x^2)$

In other words, the bad function $\left(\frac{x}{y}\right)$ is disguised as a good function $(\sqrt[3]{x})$ on Γ.

So how do we fix this? ($\frac{x}{y}$ should be in it but it is not)

Basically the problem is that $R_\Gamma = \frac{\mathbb{R}[x,y]}{\langle y^3 - x^2 \rangle}$ does not contain all the polynomial functions that it "should" have.

Informally: R_Γ is supposed to represent the space of all "nice" functions on Γ.

The fix: Let's just add back in the missing fcns and see what happens.

$\Rightarrow$ We want to add $z(x,y) = \frac{x}{y}$ to $R_\Gamma = \frac{\mathbb{R}[x,y]}{\langle y^3 - x^2 \rangle}$

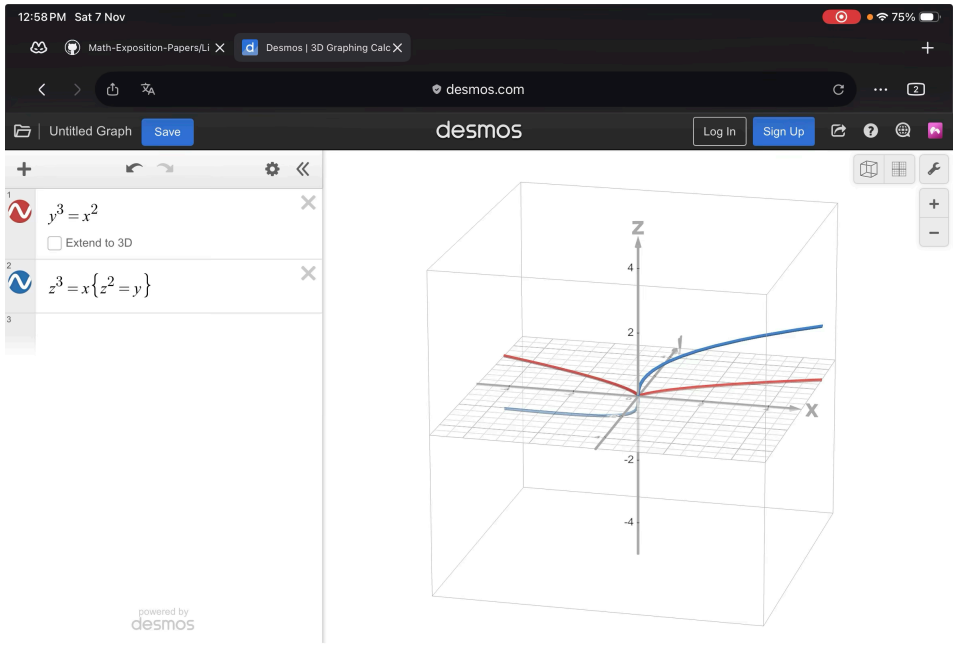
$\Rightarrow$ We want $R_\Gamma\left[\frac{x}{y}\right]$ ← "adding $\frac{x}{y}$ to R_Γ"

$$R_\Gamma\left[\frac{x}{y}\right] \cong \frac{\frac{\mathbb{R}[x,y]}{\langle y^3 - x^2 \rangle}[z]}{\langle yz - x \rangle}$$

$$\cong \frac{\mathbb{R}[x,y,z]}{\langle z^3 - x,\ z^2 - y \rangle}$$

(you can check this yourself/just take this for granted)

↑ What is this?



This is now a smooth curve.

$\Rightarrow$ We have solved the singular point issue by doing only algebra.

As a final remark, notice that the new curve P' has a canonical projection $P' \twoheadrightarrow P$ by forgetting the z-axis.

Notice that this projection corresponds ring-theoretically to the embedding $\frac{\mathbb{R}[x,y]}{\langle y^2 - x^3 \rangle} \hookrightarrow \frac{\mathbb{R}[x,y,z]}{\langle z^3 - x, z^2 - y \rangle}$. #

Things you must know: commutative algebra

Things you should know before doing AG: (strongly suggested)

- Complex analysis (behavior of holomorphic fctns & Riemann surfaces)
- Differential geometry (differentials, tangent spaces)
- Algebraic topology (sheaf cohomology)

these help with intuition massively (otherwise AG will look like ring theory slop)

what that area gives intuition for

(in descending order of importance)

Example: $\mathcal{O}(\mathbb{P}^1_k) = k$

↳ Intuition comes from complex analysis, namely any bounded holomorphic fctn is constant (Liouville's thm)

Example / def?: A variety of dimension n is said to be nonsingular at a point p if $\dim_{\kappa(p)} \frac{m_p}{m_p^2} = n$

↳ Intuition: For a smooth manifold X, the $\mathbb{R}$-vector space $\frac{m_p}{m_p^2}$ is exactly the space of all differentials at that point

A pseudo-application of why the "every ring is a ring of fctns" perspective is interesting:

Recall from complex analysis: If f is meromorphic on $\mathbb{C}$, then there $\exists$ g, h holomorphic s.t. $f = \frac{g}{h}$.

$\rightsquigarrow K(\mathbb{C}) = \mathrm{Frac}(\mathcal{O}(\mathbb{C}))$

$K(\mathbb{C})$: field of meromorphic functions; $\mathcal{O}(\mathbb{C})$: ring of holomorphic functions

Notice: $\mathbb{Q} = \mathrm{Frac}(\mathbb{Z})$ $\rightsquigarrow$ can we make an analogy of that here?

In AG (specifically arithmetic geometry $\subset$ AG), because of the discussion of the talk, we think of elements of $\mathbb{Z}$ as functions on the space $\mathrm{Spec}\,\mathbb{Z}$.

$\Rightarrow$ Rational numbers can be thought of as "meromorphic functions" on the space $\mathrm{Spec}\,\mathbb{Z}$.

eg. $\frac{5}{6}$ is the "meromorphic function" with
- a double zero at (5)
- a simple pole at (2)
- a simple pole at (3)

Intuition: Complex analysis: $\forall f$ holomorphic on h, (Weierstrauss factorization theorem) we may write $h = (z-z_0)E_{z_0}(z) \cdot e^g$ where z_0 are all the zeros of h, E_{z_0} are correction factors s.t. $(z-z_0)E_{z_0}(z)$ & e^h is holomorphic